



ICD-O-3

Carcinoma, urothelial NOS
8/20/13

Site: Bladder NOS
C67.9

QW 8/20/13

Anatomic Site: Right Ureter (FS)
Left Ureter (FS)
Bladder (FS on urethra)
Distal right ureter (FS)
Right pelvic lymph nodes (FS)
Final urethral margin
Left pelvic lymph nodes
Final left ureter

Clinical Diagnosis/History:

The patient is a ~~4~~-year-old woman. No further clinical history is provided. A recent consult report indicates a transurethral bladder tumor resection showing invasive high-grade urothelial carcinoma with invasion of the muscularis propria.

Gross Description:

The specimen is received in eight parts, each labeled with the patient's name and medical record number. Parts A through E are received fresh. Parts F-H are received in formalin.

Part A is additionally labeled "right ureter (frozen)." It consists of a single segment of ureter, measuring 2.8 cm in length and 1.2 cm in greatest diameter. The specimen is oriented by a black stitch and verbally by the surgeon. Gross tumor extends from the distal end of the ureter. The proximal margin is submitted en face for frozen section diagnosis as FS1, with the frozen section remnant submitted in cassette A1. The remaining ureter is entirely submitted as follows:

Cassette A2: Proximal half of remaining ureter.
Cassette A3: Distal half of remaining ureter.

Part B is additionally labeled "left ureter (frozen)." It consists of one fragment of pink-red ureter, measuring 0.5 x 0.4 x 0.3 cm. The specimen is entirely submitted for frozen section diagnosis as FS2, with the frozen section remnant submitted in cassette B1.

Part C is additionally labeled "bladder." It consists of a specimen, weighing 271 gm and measuring 22 x 13 x 3.5 cm. Present within the specimen is a bladder, with a stent protruding through the left ureter. In addition, there is included the left fallopian tube and ovary. There is a small portion of the vaginal cuff posterior to the urethra. The small stub of the left ureter measures 0.3 cm in length and 0.2 cm in diameter and is grossly normal.

The remnant fragment of right ureter measures 1.5 cm in length and 0.9 cm in diameter, and there is gross tumor at the proximal end. An en face margin of the urethra is submitted for frozen section diagnosis as FS3, with the frozen section remnant submitted in cassette C1. The right-anterior

half of the bladder is inked green, the left-anterior half is inked blue, and the posterior is inked black. The specimen is then opened anteriorly through the remaining urethra to reveal a yellow-tan, friable mass, measuring 3 x 2 x 1.8 cm, near the right ureter opening and posterior bladder wall. Within the posterior aspect, there is an area of potential bladder wall involvement by the tumor, with possible tumor within 0.1 cm of the posterior surgical resection margin. No other focal lesions are identified. The left fallopian tube measures 5.5 cm in length and 0.4 cm in average diameter. The fimbriae are lush, and cross sections are grossly normal. The left ovary measures 2.5 x 1.2 x 0.9 cm and has a grossly normal parenchyma. The perivesicular fat is extensively searched for lymph nodes, with two candidate lymph nodes identified. Representative sections are submitted as follows:

Cassette C1:	Remnant from FS3.
Cassette C2:	Right and left ureter margins.
Cassette C3:	Left fallopian tube and ovary.
Cassette C4:	Normal bladder.
Cassette C5:	Tumor at posterior bladder.
Cassette C6:	Tumor at posterior bladder.
Cassette C7:	Mass at right ureter.
Cassette C8:	Mass at candidate remnant urethra.
Cassette C9:	Two candidate lymph nodes and candidate right fallopian tube.

Part D is additionally labeled "distal right ureter, long stitch = proximal end." It consists of a single red-tan, tubular structure, measuring 4.3 cm in length and 1.4 cm in greatest diameter. The lumen measures 0.4 cm in diameter. A long stitch marks the proximal margin, and a short stitch marks the distal margin. An en face cut of the proximal margin is submitted for frozen section diagnosis as FS4, with the frozen section remnant submitted in cassette D1. The remaining specimen is entirely submitted as follows:

Cassette D2:	Remaining proximal one-third.
Cassette D3:	Remaining mid one-third.
Cassette D4:	Remaining distal one-third.

Part E, additionally labeled "right pelvic lymph nodes," consists of soft, red-yellow, fibro-fatty tissue measuring 8 x 5.5 x 1 cm and weighing 26 gm. Multiple prominent lymph node candidates are identified. Representative sections from two lymph node candidates are submitted for frozen section diagnosis 5. The first frozen section remnant is submitted in cassette E1, with the remainder of the lymph node candidate submitted in cassette E2. The second frozen section remnant is submitted in cassette E3, with the remainder of the lymph node candidates submitted in cassettes E4-E5. Following fixation, the remainder of the lymph node candidates are submitted as follows:

Cassette E6:	One lymph node, bisected.
Cassette E7:	One lymph node candidate, bisected.
Cassette E8:	One lymph node candidate, bisected.
Cassette E9:	Two lymph node candidates, bisected and differentially inked.

Cassette E10:	Three small lymph node candidates, bisected, two differentially inked.
Cassette E11:	Multiple small lymph node candidates, submitted whole.

Part F, additionally labeled "4 - final urethral margin," consists of one

firm, yellow-brown, irregular, unoriented tissue fragment measuring 2.2 x 1.8 x 0.4 cm. The specimen is sectioned and entirely submitted in cassette F1.

Part G, additionally labeled "7 - left pelvic lymph nodes," consists of multiple soft, yellow, irregular, unoriented, fibroadipose tissue fragments measuring 9 x 5 x 3.5 cm in aggregate and containing multiple lymph node candidates ranging from 0.3 to 3.1 cm in greatest diameter. The lymph node candidates are submitted as follows:

Cassette G1: Intact lymph node candidates.
Cassette G2: Two bisected lymph node candidates, inked in different colors.
Cassettes G3-G5: Single lymph node candidate, trisected.
Cassettes G6-G9: Single lymph node candidate, sectioned into four sections.

Part H, additionally labeled "8 - final left ureter," consists of one soft, tan-white, irregular, unoriented, fibrous tissue fragment measuring 1 x 0.7 x 0.2 cm. The specimen is trisected and entirely submitted in cassette H1.

The immunoperoxidase stain(s) reported above were developed and their performance characteristics determined by the

They have not been cleared or approved by the U. S. Food and Drug Administration. The FDA has determined that such clearance or approval is not necessary. These tests are used for clinical purposes. They should not be regarded as investigational or for research. This laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 ("CLIA") as qualified to perform high-complexity clinical testing.

Diagnosis Comments:

Bladder Tumor Synoptic Comment

1. Tumor type: Urothelial (transitional cell).
2. Tumor grade: High grade.
3. Tumor size: 3 cm.
4. Extent of tumor in bladder: Invasion into outer half of muscularis propria.
5. Lymphatic/vascular invasion: None identified.
6. Epithelial abnormalities in bladder: Carcinoma in situ.
7. Extension of tumor into organs adjacent to bladder: Ureter, right.
8. Surgical margins:
 - Urethral margin: Free of tumor.
 - Right ureter margin: Final margin free of tumor.
 - Left ureter margin: Final margin free of tumor.
 - Perivesical margin: Negative, closest margin is 0.2 cm, posterior, slide

C5.

9. Lymph nodes: No tumor (0/24).
10. Other pathologic findings in bladder: Ulceration and chronic inflammation.
11. AJCC/UICC stage: pT2bN0MX.
12. Additional margin: The distal end of Part A (right ureter) is involved by high grade invasive carcinoma (slide A3) into the muscularis propria; however the proximal specimen margin (slide A1) demonstrates chronic inflammation of the lamina propria with atypia of the urothelial mucosa with we regard as reactive atypia. In Part D (labeled final proximal margin of the right ureter) the proximal specimen margin (slide D1) is negative for dysplasia or carcinoma. However the distal end of this specimen

(slide D4) reveals nuclear atypia of urothelial carcinoma in situ. Because of the intense chronic inflammation, distinguishing reactive vs dysplastic changes are difficult. Immunohistochemical stain for p53 was performed on slide D3 where the inflamed urothelium have atypical, round, slightly enlarged nuclei and prominent nucleoli. The stains highlights scattered positivity. Thus based on the morphology and results of the immunohistochemical stain, the atypical changes on slide D3 are regarded as reactive. Dr. [REDACTED] has reviewed select portions of Parts A and D and concurs. Level section on C9 confirms the diagnosis.

Interoperative Diagnosis:

FS1 (A) Right ureter, proximal margin, biopsy: Chronic ureteritis with atypia, favor dysplasia. [REDACTED]

FS2 (B) Left ureter, biopsy: No tumor seen. [REDACTED]

FS3 (C) Distal urethra, biopsy: No tumor seen. [REDACTED]

FS4 (D) Right ureter, proximal margin, biopsy: No tumor seen. [REDACTED]

FS5 (E) Right pelvic lymph nodes, biopsy: Representative section of two lymph node candidates, no tumor seen. [REDACTED]

Final Diagnosis:

A. Right ureter, distal, resection:

1. Proximal specimen margin with chronic ureteritis and atypia; see comment.
2. Distal end of ureter with high grade invasive urothelial carcinoma into muscularis propria.

B. Left ureter, distal, biopsy: No dysplasia or carcinoma.

C. Bladder and left fallopian tube and ovary, cystectomy and salpingo-oophorectomy:

1. Bladder with invasive high grade urothelial carcinoma, into outer one-half of muscularis propria; see comment.
2. No tumor in one lymph node (0/1).
3. Left fallopian tube and ovary with no significant pathologic abnormality.

D. Right ureter, final proximal margin, biopsy: Proximal specimen margin with no dysplasia or carcinoma.

E. Lymph nodes, right pelvic, dissection: No tumor in seventeen lymph nodes (0/17).

F. Urethra, "final margin," biopsy: No dysplasia or carcinoma.

G. Lymph nodes, left pelvic, dissection: No tumor in six lymph nodes (0/6).

H. Left ureter, final margin, biopsy: No dysplasia or carcinoma.

Criteria	Yes	No
Diagnosis Discrepancy		☒
Primary Tumor Site Discrepancy		☒
HIPAA Discrepancy		☒
Prior Malignancy History		☒
Dual/Synchronous Primary Noted		☒
Case is (circle):	QUALIFIED / DISQUALIFIED	
Reviewer Initial	Date Reviewed: 8/9/13	